

Calibration of an ESP32 Soil-Moisture Sensor

Using Controlled Water Addition

Name: Chan Yunchen

Date: 12 August 2026

1. Research Question

How does adding 0 g, 20 g, and 40 g of water affect the ADC readings of an ESP32 soil-moisture sensor? Can an ADC threshold of 2200 distinguish dry soil from watered soil?

2. Hypothesis

As the amount of added water increases, the ADC reading will decrease. The dry sample is expected to have an ADC reading above 2200, while the watered samples are expected to have readings below 2200.

3. Materials

- One ESP32 development board
- One resistive soil-moisture sensor
- A laptop with Thonny and MicroPython
- A USB cable and jumper wires
- Three identical plastic cups
- The same type and amount of dry growing medium
- Water
- A device for measuring water mass
- A ruler
- Labels and a marker

4. Method

1. Three identical plastic cups were filled with approximately the same amount and height of dry growing medium.
2. The cups were labelled A, B, and C. Cup A received 0 g of water, Cup B received 20 g, and Cup C received 40 g.
3. The growing medium in each cup was mixed, and the samples were left for approximately 10 minutes to allow the water to spread.

4. The soil-moisture sensor was connected to the ESP32, and a MicroPython program was run using Thonny.
5. The sensor was inserted vertically to a depth of 4 cm. The same sensor and insertion depth were used for every measurement.
6. The program collected five rapid ADC readings and calculated their average to produce each displayed ADC value.
7. For each cup, the sensor was placed at five different positions. One stable displayed ADC value was recorded at each position.
8. This produced five formal measurements for each water condition and 15 measurements in total.
9. During initial testing, a loose signal wire produced constant readings of 4095. The wire was reconnected, and these invalid readings were excluded before formal data collection began.

5. Results

Table 1. ADC readings under different water-addition conditions

Condition	Water added (g)	ADC readings	Mean ADC	Min–max	Range	Sample SD
A	0	3057, 3620, 3220, 3393, 3079	3273.8	3057– 3620	563	235.5
B	20	1890, 1904, 1923, 1825, 1806	1869.6	1806– 1923	117	51.2
C	40	1102, 1037, 1208, 1179, 1089	1123.0	1037– 1208	171	69.6

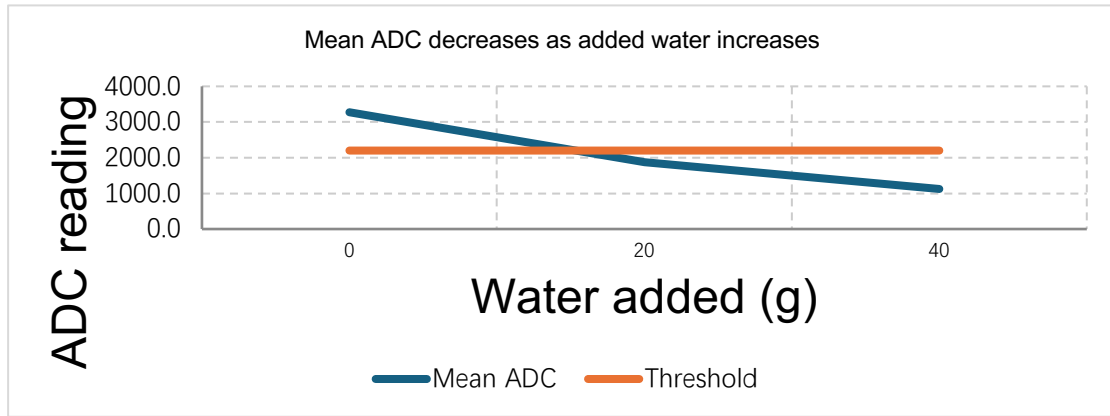


Figure 1. Relationship between added water mass and mean ADC reading. The orange line represents the watering threshold of 2200 ADC.

Results Description

The mean ADC reading decreased as the amount of added water increased. Cup A, with 0 g of added water, had the highest mean ADC reading of 3273.8 ± 235.5 . Cup B, with 20 g of water, had a mean reading of 1869.6 ± 51.2 . Cup C, with 40 g of water, had the lowest mean reading of 1123.0 ± 69.6 .

All five readings from Cup A were above the watering threshold of 2200, whereas all readings from Cups B and C were below the threshold.

6. Discussion

The results support the hypothesis that the ADC reading decreases as the amount of water increases. The mean ADC value decreased from 3273.8 at 0 g of water to 1869.6 at 20 g and 1123.0 at 40 g. This shows that the sensor was able to detect differences in the water content of the growing medium.

The threshold of 2200 successfully separated the dry condition from the two watered conditions in this experiment. Cup A produced readings above 2200 and would activate the water pump, while Cups B and C produced readings below 2200 and would not activate the pump. Therefore, 2200 appears to be a suitable preliminary threshold for this particular growing medium and experimental setup.

Cup A had the largest standard deviation, indicating that measurements in the dry growing medium were less consistent. This may have been caused by differences in sensor contact, insertion position, or the uneven structure of the growing medium.

7. Limitations

This experiment had several limitations. First, only one cup was prepared for each water condition. The five readings were taken at different positions within the same cup, so they were repeated measurements rather than five independently prepared samples.

Second, the growing medium may not have been completely uniform. Differences in water distribution, sensor contact, and insertion position could have affected the ADC readings. Although the sensor was inserted to the same depth of 4 cm, its exact angle and surrounding material may have varied slightly.

Third, only one soil-moisture sensor, three water levels, and one type of growing medium were tested. Therefore, the results cannot yet be applied to every soil type or plant. More independently prepared samples and long-term testing would improve the reliability of the results.

During initial testing, a loose signal wire caused invalid readings of 4095. These readings were excluded because normal variable readings returned after the wire was reconnected.

8. Conclusion

The experiment showed that the ADC reading decreased as the amount of added water increased. The mean ADC readings were 3273.8 for 0 g, 1869.6 for 20 g, and 1123.0 for 40 g of water. Therefore, the results supported the hypothesis.

An ADC threshold of 2200 successfully distinguished the dry sample from the two watered samples under the conditions of this experiment. However, further testing with independently prepared samples, different growing media, and longer measurement periods is required before this threshold can be used more widely.

Appendix A. Experimental Evidence



Figure 2. Three growing-medium samples prepared with 0 g, 20 g, and 40 g of added water.



Figure 3. The soil-moisture sensor was marked to maintain an insertion depth of approximately 4 cm.

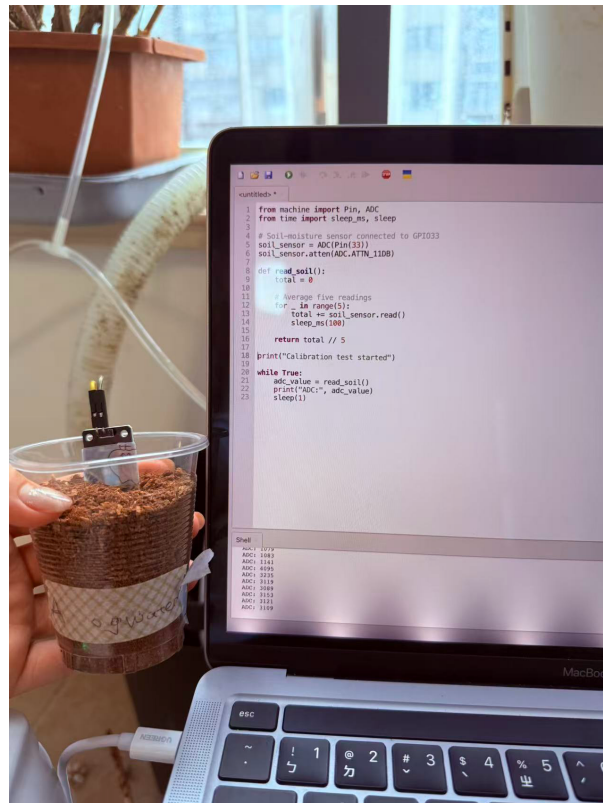


Figure 4. ADC measurement for Cup A with 0 g of added water.

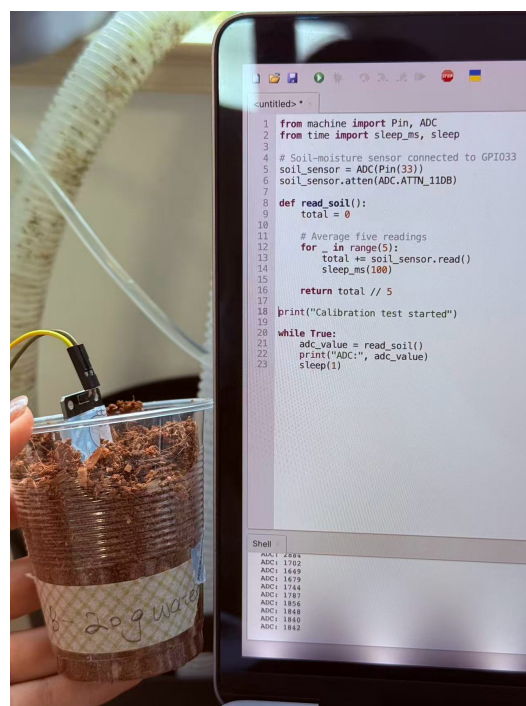


Figure 5. ADC measurement for Cup B with 20 g of added water.

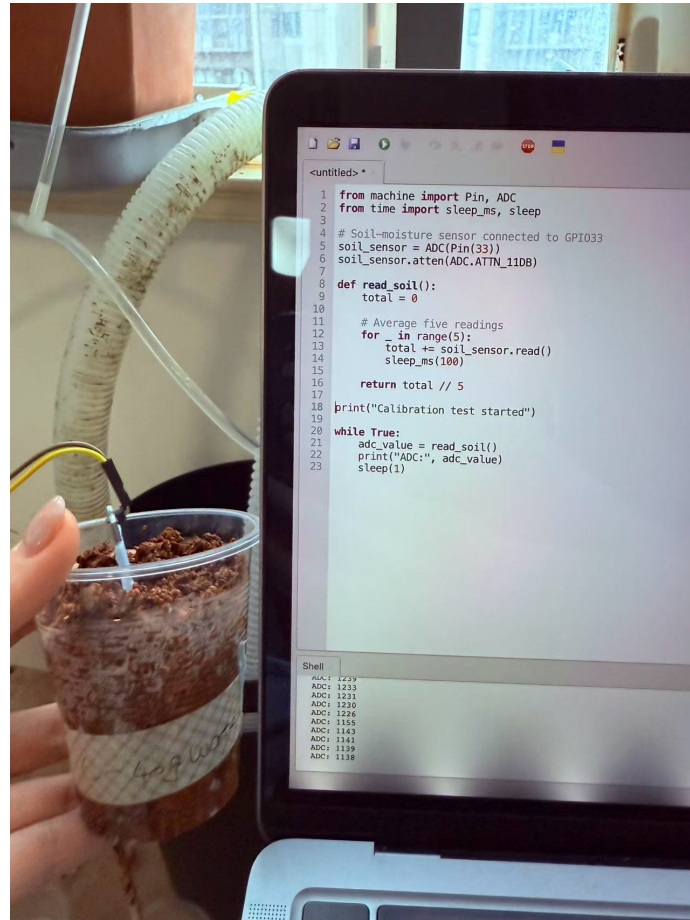


Figure 6. ADC measurement for Cup C with 40 g of added water.

Appendix B. Calibration Code

The following MicroPython code was used to read the soil-moisture sensor through GPIO33. Each displayed ADC value was calculated as the average of five rapid readings taken at 100 ms intervals.

```
from machine import Pin, ADC
from time import sleep_ms, sleep

soil_sensor = ADC(Pin(33))
soil_sensor.atten(ADC.ATTN_11DB)

def read_soil():
    total = 0

    for _ in range(5):
        total += soil_sensor.read()
        sleep_ms(100)

    return total // 5

print("Calibration test started")

while True:
    adc_value = read_soil()
    print("ADC:", adc_value)
    sleep(1)
```

The complete source code and project documentation are available in the following GitHub repository:

<https://github.com/yunchen980604-dev/Sensor-Driven-Smart-Watering-System>